

Loading Libraries ---

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

Loading Dataset ---

```
df = pd.read_csv("Telco-Customer-Churn.csv")
```

Basic Checks on the Imported Data ---

1. Data Preview ---

```
df.head()
```

	customerID	gender	SeniorCitizen	Partner	Dependents	tenure
0	7590-VHVEG	Female	0	Yes	No	1
1	5575-GNVDE	Male	0	No	No	34
2	3668-QPYBK	Male	0	No	No	2
3	7795-CF0CW	Male	0	No	No	45
4	9237-HQITU	Female	0	No	No	2

	MultipleLines	InternetService	OnlineSecurity	...
0	No phone service	DSL	No	...
1	No	DSL	Yes	...
2	No	DSL	Yes	...
3	No phone service	DSL	Yes	...
4	No	Fiber optic	No	...

	TechSupport	StreamingTV	StreamingMovies	Contract
0	No	No	No	Month-to-month
1	No	No	No	One year

2	No	No	No	Month-to-month
Yes				
3	Yes	No	No	One year
No				
4	No	No	No	Month-to-month
Yes				

	PaymentMethod	MonthlyCharges	TotalCharges	Churn
0	Electronic check	29.85	29.85	No
1	Mailed check	56.95	1889.5	No
2	Mailed check	53.85	108.15	Yes
3	Bank transfer (automatic)	42.30	1840.75	No
4	Electronic check	70.70	151.65	Yes

[5 rows x 21 columns]

1. Metadata ---

```
df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7043 entries, 0 to 7042
Data columns (total 21 columns):
#   Column                Non-Null Count  Dtype
---  -
0   customerID            7043 non-null   object
1   gender                 7043 non-null   object
2   SeniorCitizen          7043 non-null   int64
3   Partner                7043 non-null   object
4   Dependents             7043 non-null   object
5   tenure                 7043 non-null   int64
6   PhoneService           7043 non-null   object
7   MultipleLines           7043 non-null   object
8   InternetService         7043 non-null   object
9   OnlineSecurity          7043 non-null   object
10  OnlineBackup            7043 non-null   object
11  DeviceProtection        7043 non-null   object
12  TechSupport             7043 non-null   object
13  StreamingTV             7043 non-null   object
14  StreamingMovies         7043 non-null   object
15  Contract                7043 non-null   object
16  PaperlessBilling        7043 non-null   object
17  PaymentMethod           7043 non-null   object
18  MonthlyCharges          7043 non-null   float64
19  TotalCharges            7043 non-null   object
20  Churn                   7043 non-null   object
dtypes: float64(1), int64(2), object(18)
memory usage: 1.1+ MB
```

1. Stats ---

```
df.describe()
```

	SeniorCitizen	tenure	MonthlyCharges
count	7043.000000	7043.000000	7043.000000
mean	0.162147	32.371149	64.761692
std	0.368612	24.559481	30.090047
min	0.000000	0.000000	18.250000
25%	0.000000	9.000000	35.500000
50%	0.000000	29.000000	70.350000
75%	0.000000	55.000000	89.850000
max	1.000000	72.000000	118.750000

1. Counting Nulls

```
df.isnull().sum()
```

customerID	0
gender	0
SeniorCitizen	0
Partner	0
Dependents	0
tenure	0
PhoneService	0
MultipleLines	0
InternetService	0
OnlineSecurity	0
OnlineBackup	0
DeviceProtection	0
TechSupport	0
StreamingTV	0
StreamingMovies	0
Contract	0
PaperlessBilling	0
PaymentMethod	0
MonthlyCharges	0
TotalCharges	0
Churn	0

dtype: int64

1. Dimensions ---

```
df.shape
```

```
(7043, 21)
```

Data Cleaning ---

Missing Values ---

```
print(df['TotalCharges'].dtype)
```

object

1. Replacing Blank Spaces---

```
df['TotalCharges'] = df['TotalCharges'].replace(' ', np.nan)
```

1. Converting data to numeric ---

```
df['TotalCharges'] = pd.to_numeric(df['TotalCharges'],  
errors='coerce')
```

1. Verifying the Data Type again ---

```
print(df['TotalCharges'].dtype)  
float64
```

1. Checking Missing Values ---

```
print(df['TotalCharges'].isnull().sum())  
11
```

1. Filling these Missing Values ---

```
df['TotalCharges'] =  
df['TotalCharges'].fillna(df['TotalCharges'].median())
```

1. Checking the Missing Values again ---

```
print(df['TotalCharges'].isnull().sum())  
0
```

Checking for Duplicates ---

```
df.duplicated().sum()  
np.int64(0)
```

Checking for Unique Values ---

```
for col in df.columns:  
    print(f"\n{col}")  
    print(df[col].unique())  
  
customerID  
['7590-VHVEG' '5575-GNVDE' '3668-QPYBK' ... '4801-JZAZL' '8361-LTMKD'  
 '3186-AJIEK']
```

```
gender
['Female' 'Male']

SeniorCitizen
[0 1]

Partner
['Yes' 'No']

Dependents
['No' 'Yes']

tenure
[ 1 34  2 45  8 22 10 28 62 13 16 58 49 25 69 52 71 21 12 30 47 72 17
27
 5 46 11 70 63 43 15 60 18 66  9  3 31 50 64 56  7 42 35 48 29 65 38
68
32 55 37 36 41  6  4 33 67 23 57 61 14 20 53 40 59 24 44 19 54 51 26
0
39]
```

```
PhoneService
['No' 'Yes']

MultipleLines
['No phone service' 'No' 'Yes']

InternetService
['DSL' 'Fiber optic' 'No']

OnlineSecurity
['No' 'Yes' 'No internet service']

OnlineBackup
['Yes' 'No' 'No internet service']

DeviceProtection
['No' 'Yes' 'No internet service']

TechSupport
['No' 'Yes' 'No internet service']

StreamingTV
['No' 'Yes' 'No internet service']

StreamingMovies
['No' 'Yes' 'No internet service']

Contract
['Month-to-month' 'One year' 'Two year']
```

```
PaperlessBilling
['Yes' 'No']

PaymentMethod
['Electronic check' 'Mailed check' 'Bank transfer (automatic)'
 'Credit card (automatic)']

MonthlyCharges
[29.85 56.95 53.85 ... 63.1 44.2 78.7 ]

TotalCharges
[ 29.85 1889.5 108.15 ... 346.45 306.6 6844.5 ]

Churn
['No' 'Yes']

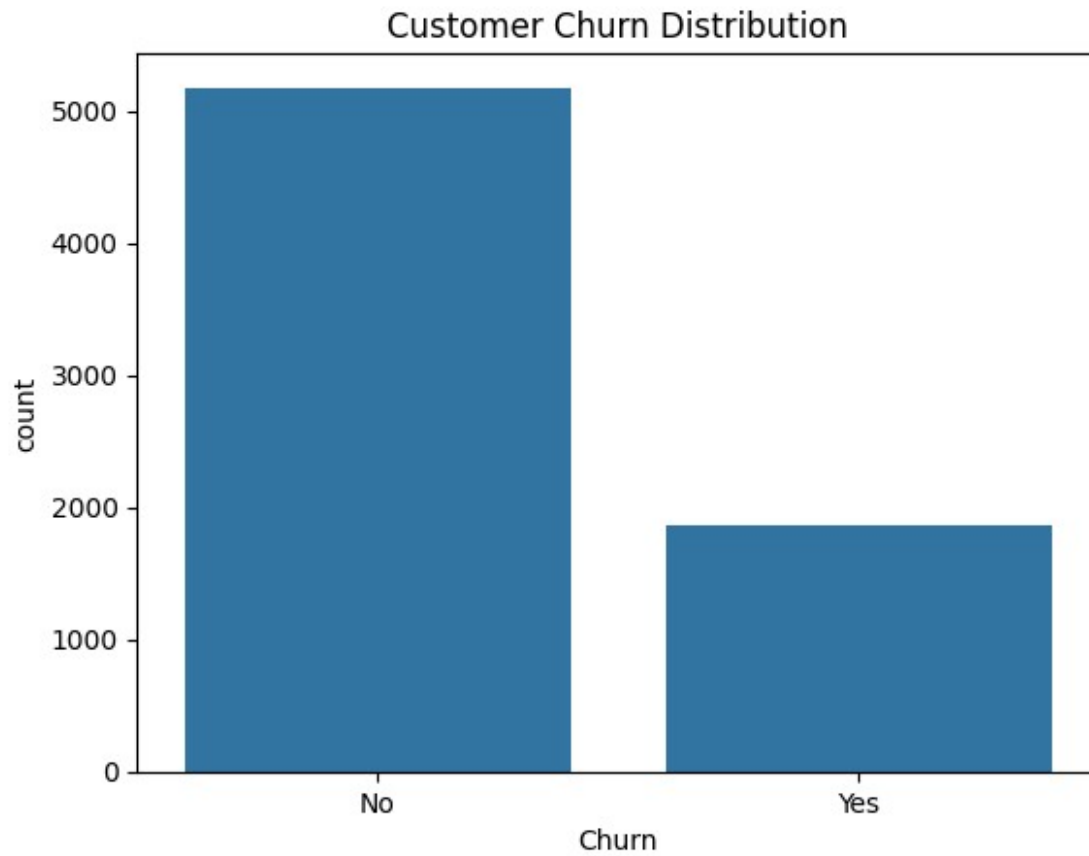
Exporting the Cleaned Dataset ---

df.to_csv(
    "telco_cleaned.csv",
    index=False
)
```

Exploratory Data Analysis (EDA) ---

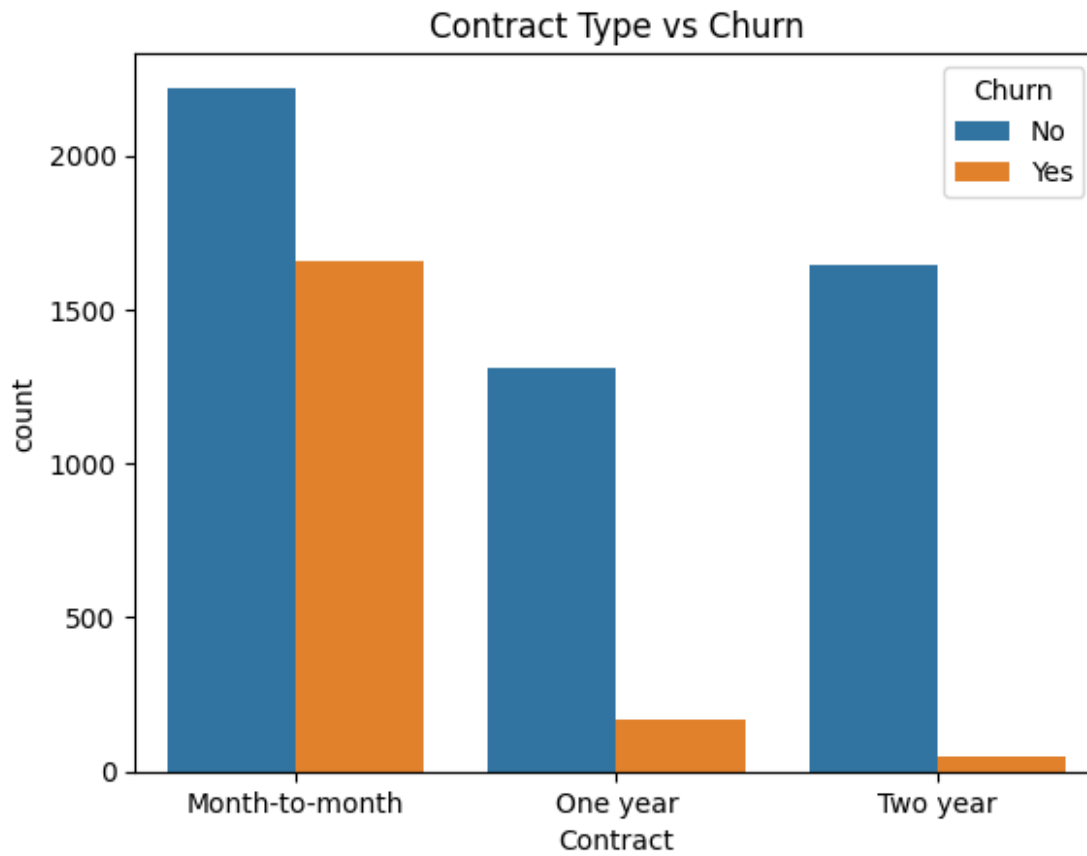
1. Churn Distribution ---

```
sns.countplot(x='Churn', data=df)
plt.title('Customer Churn Distribution')
plt.show()
```



1. Contract Type vs Churn

```
sns.countplot(  
    x='Contract',  
    hue='Churn',  
    data=df  
)  
plt.title('Contract Type vs Churn')  
plt.show()
```

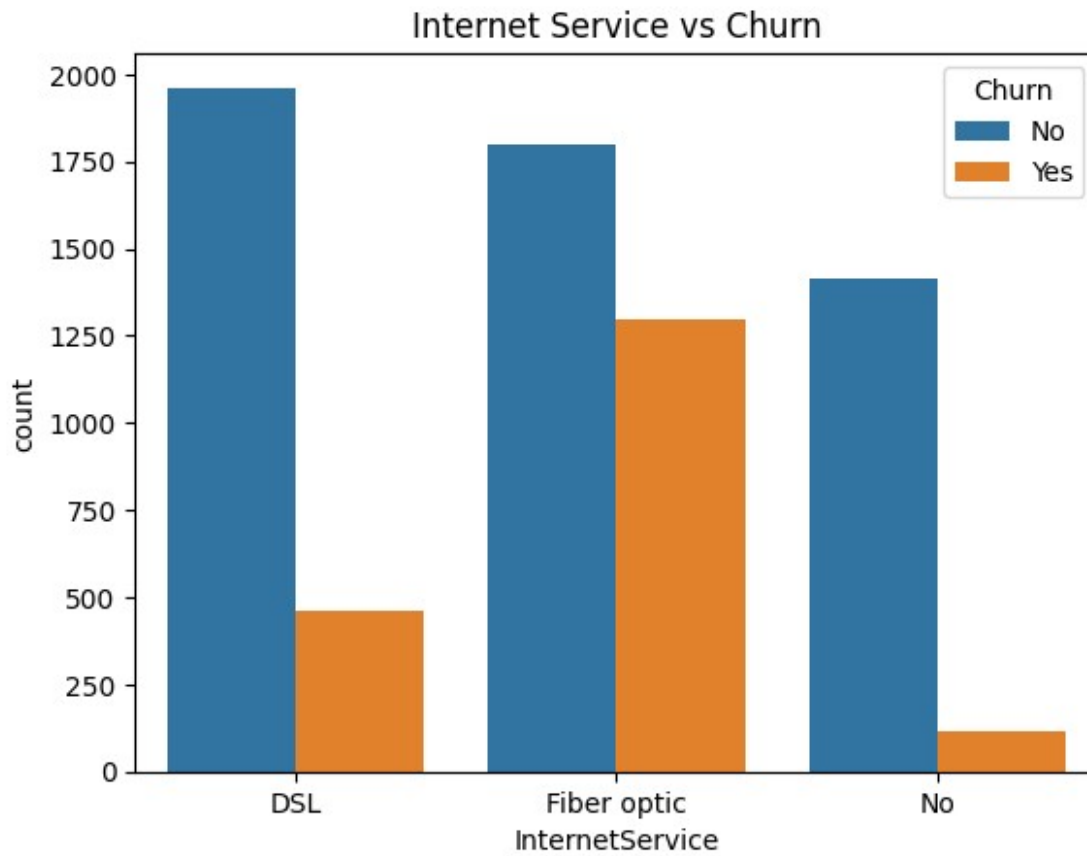


2.1 Observation: Customers with Month-to-Month contracts show the highest churn. Customers with One-Year and Two-Year contracts have significantly lower churn rates. Churn decreases as contract duration increases.

2.2 Business implication: Long-term contracts improve customer retention. Customers without a long-term commitment are more likely to switch providers. The company should encourage Month-to-Month customers to move to annual plans through discounts and loyalty benefits.

1. Internet Service vs Churn

```
sns.countplot(
    x='InternetService',
    hue='Churn',
    data=df
)
plt.title('Internet Service vs Churn')
plt.show()
```

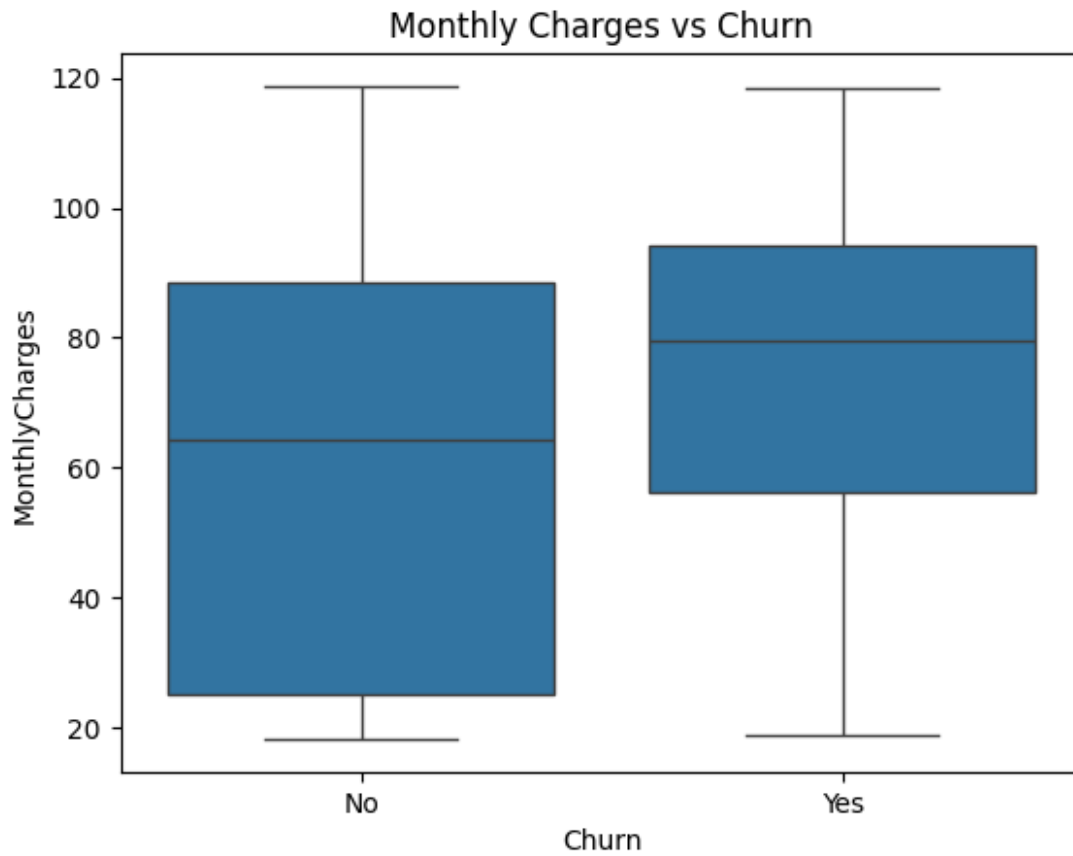



3.1 Observation: Customers using Fiber Optic Internet exhibit the highest churn. DSL customers have comparatively lower churn. Customers without internet service show the lowest churn.

3.2 Business Implication: Despite being a premium service, Fiber Optic customers are leaving more frequently. Possible reasons include higher pricing, service quality issues, or unmet expectations. Customer feedback and service-quality investigations should focus on Fiber Optic users.

1. Monthly Charges vs Churn

```
sns.boxplot(  
    x='Churn',  
    y='MonthlyCharges',  
    data=df  
)  
plt.title('Monthly Charges vs Churn')  
plt.show()
```

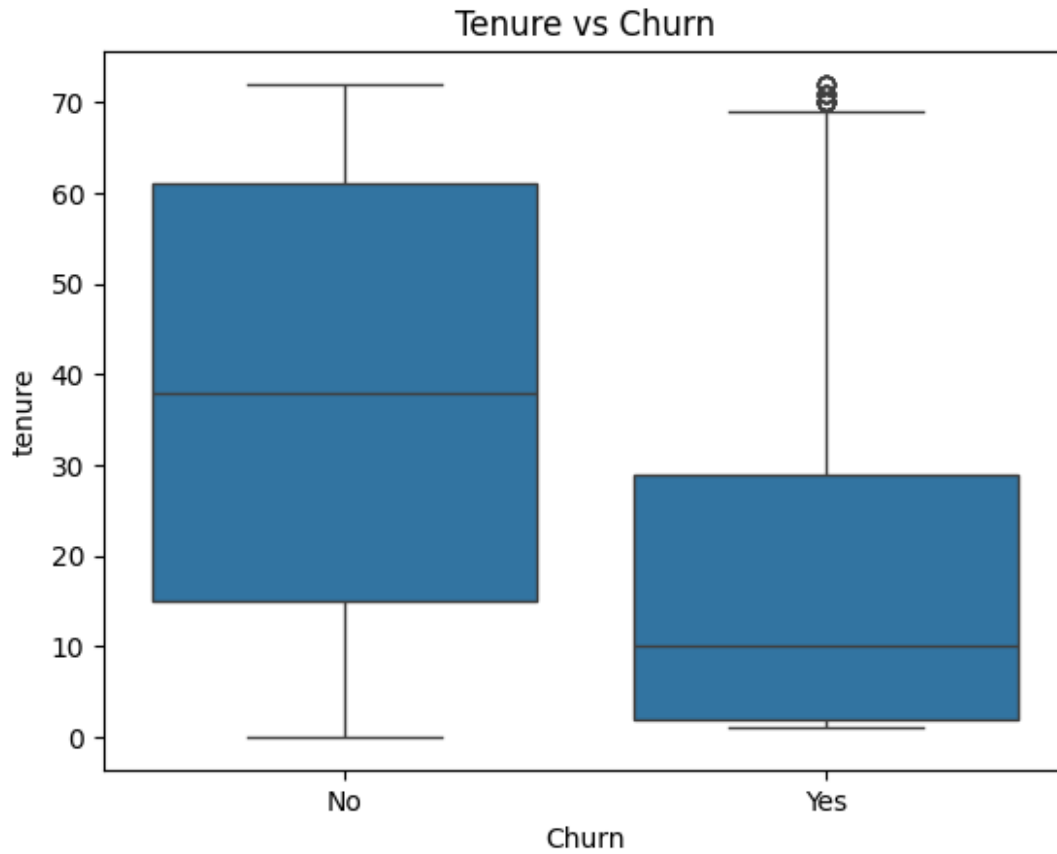


4.1 Observation: Customers who churn generally have higher monthly charges. The churned customer distribution is skewed toward expensive plans.

4.2 Business Implication: Price sensitivity may be contributing to churn. Customers paying more expect better value and service quality. The company should review premium pricing strategies and provide additional benefits to high-paying customers.

1. Tenure vs Churn

```
sns.boxplot(x='Churn', y='tenure', data=df)
plt.title('Tenure vs Churn')
Text(0.5, 1.0, 'Tenure vs Churn')
```



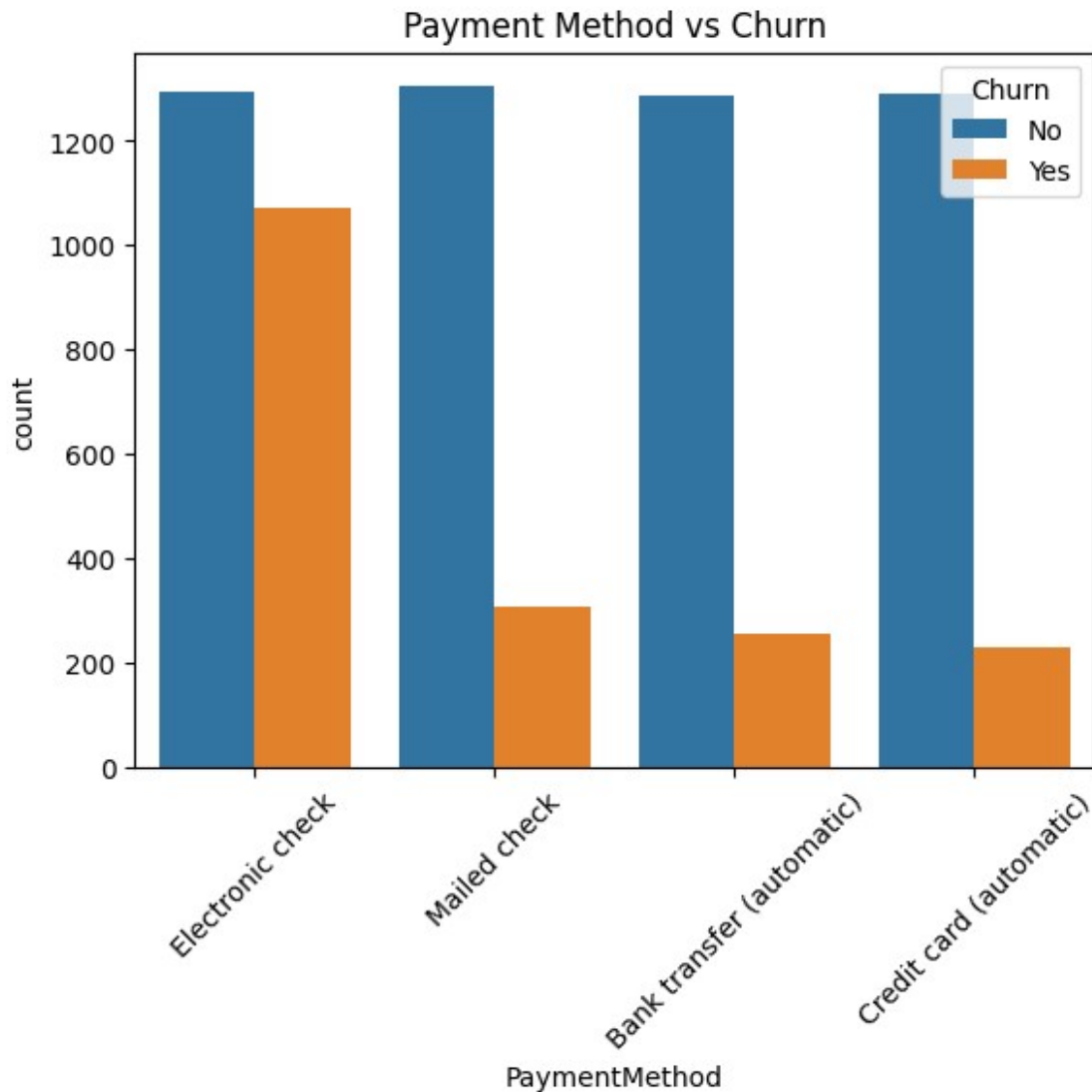
5.1 Observation: Customers with shorter tenure are much more likely to churn. Long-tenure customers generally remain loyal.

5.2 Business Implication: The first year of the customer lifecycle is critical. Onboarding and early engagement programs can significantly improve retention.

1. Payment Method vs Churn

```
sns.countplot(
    x='PaymentMethod',
    hue='Churn',
    data=df
)
plt.title('Payment Method vs Churn')
plt.xticks(rotation=45)

([0, 1, 2, 3],
 [Text(0, 0, 'Electronic check'),
  Text(1, 0, 'Mailed check'),
  Text(2, 0, 'Bank transfer (automatic)'),
  Text(3, 0, 'Credit card (automatic)')])
```



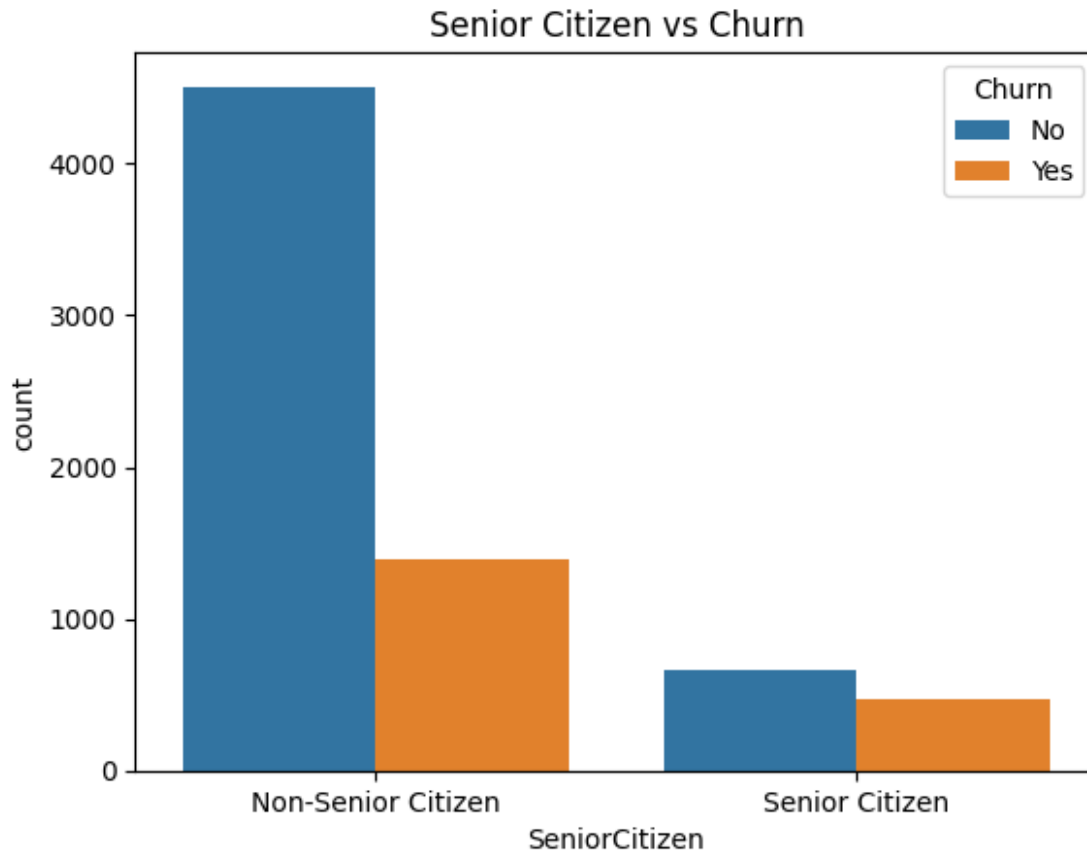
6.1 Observation: Customers using Electronic Check tend to churn more than customers using automatic payment methods.

6.2 Business Implication: Automatic payment methods indicate stronger customer commitment. Encouraging customers to enroll in auto-pay programs may improve retention.

1. Senior Citizen vs Churn

```
ax = sns.countplot(  
    x='SeniorCitizen',  
    hue='Churn',  
    data=df  
)  
  
ax.set_xticks([0, 1], ['Non-Senior Citizen', 'Senior Citizen'])
```

```
plt.title('Senior Citizen vs Churn')
plt.show()
```

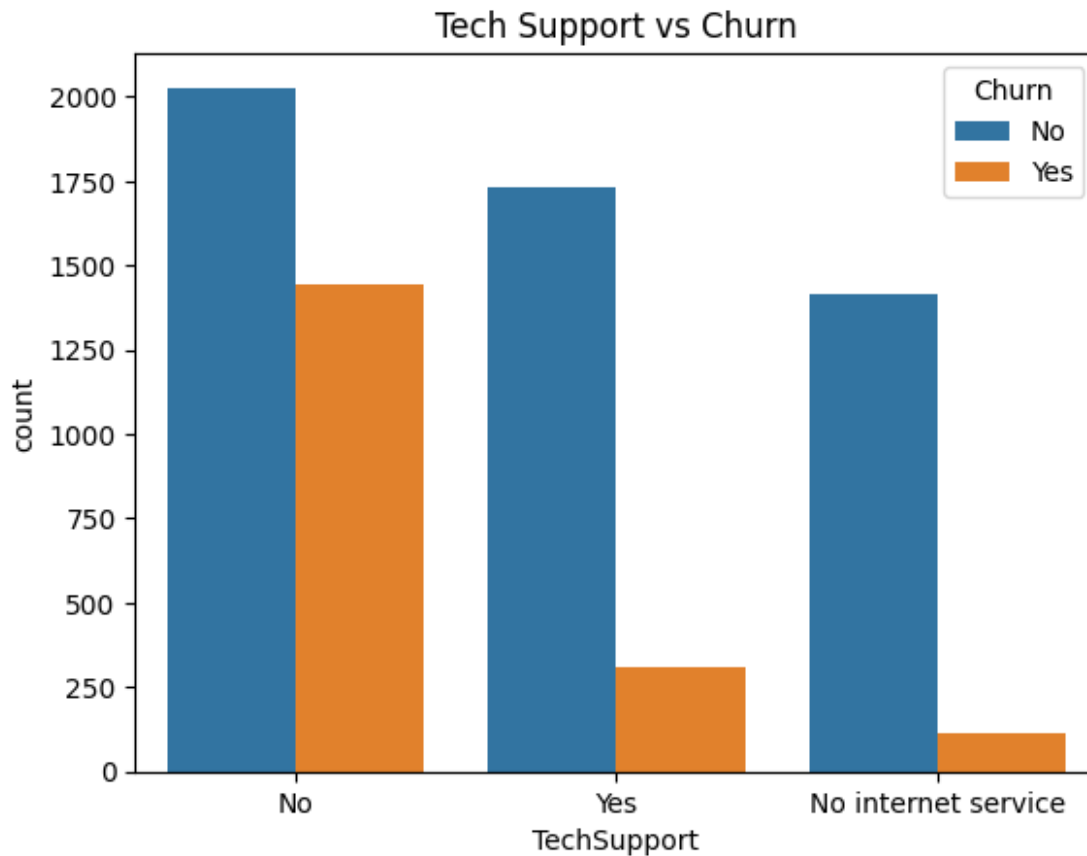


7.1 Observation: Senior citizens show a higher churn rate compared to non-senior customers.

7.2 Business Implication: Senior customers may have different service expectations or affordability concerns. Targeted support programs could help improve retention among this segment.

1. Tech Support vs Churn

```
sns.countplot(
    x='TechSupport',
    hue='Churn',
    data=df
)
plt.title('Tech Support vs Churn')
Text(0.5, 1.0, 'Tech Support vs Churn')
```

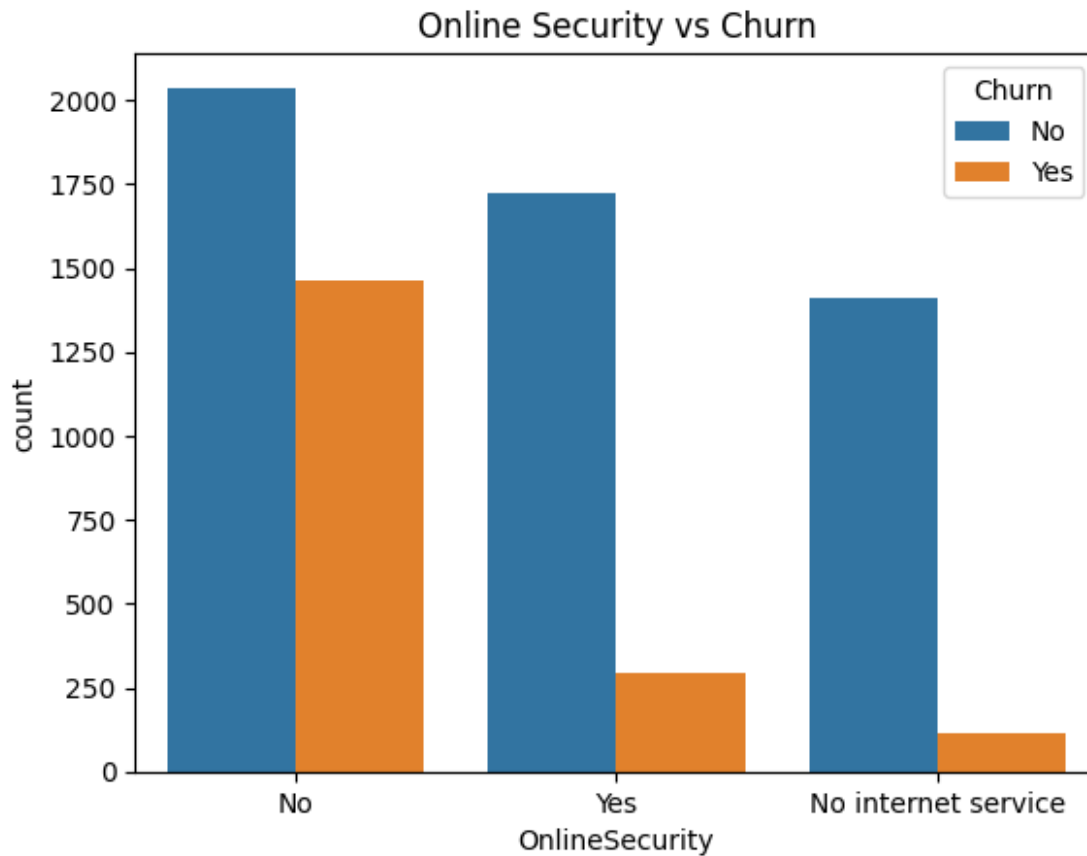


8.1 Observation: Customers without Tech Support churn significantly more than customers who subscribe to Tech Support.

8.2 Business Implication: Technical assistance contributes positively to customer satisfaction. Bundling Tech Support into service plans may reduce churn.

1. Online Security vs Churn

```
sns.countplot(
    x='OnlineSecurity',
    hue='Churn',
    data=df
)
plt.title('Online Security vs Churn')
Text(0.5, 1.0, 'Online Security vs Churn')
```



9.1 Observation: Customers without Online Security services are more likely to churn.

9.2 Business Implication: Value-added services increase customer stickiness. Promoting security packages can improve retention while increasing revenue.